

Cancer options

TL;DR

We evaluated five cancers against the five project questions (cross-dataset response prediction, cell viability modelling, ODE models, drug target discovery, and treatment type selection). The recommendation is **Melanoma (SKCM)**, with NSCLC and TNBC as credible alternatives.

🎯 Recommended: Melanoma (SKCM)

The BRAF V600E/K mutation splits patients into three clean treatment arms — dacarbazine (chemo), BRAF/MEK inhibitors (targeted), and PD-1/CTLA-4 checkpoint inhibitors — making it the strongest candidate for Q5. It also has a well-sized TCGA cohort (~473), mature ODE literature (Q3), deep DepMap coverage (~60 lines, Q2/Q4), and multiple published IO trial cohorts for cross-dataset validation (Q1). It scores well on all five questions.

Alternatives

NSCLC (LUAD + LUSC, TCGA ~1,089) is the best choice if sample size is the priority. DepMap cell line coverage is the largest of any candidate (~150+ lines), though pooling two subtypes adds biological noise.

TNBC is the strongest breast cancer option, with a compelling chemo vs. pembrolizumab + chemo story (Q5) and high clinical urgency. The TCGA cohort is small (~170) and must be supplemented with METABRIC and trial datasets; budget extra time for data harmonisation.

CRC (MSI-H subgroup) and **ccRCC** are viable but each carry a structural weakness — an extremely small immunotherapy-relevant cohort (~90–125 MSI-H patients) for CRC, and thin DepMap cell line coverage (~25–30 lines) for ccRCC.

Rejected

GBM, pancreatic cancer, AML, HER2+ breast, ovarian cancer, and undifferentiated all-subtype breast cancer were ruled out due to small cohorts, weak immunotherapy stories, divergent biology, or limited DepMap representation.

Bottom Line

Pick Melanoma. If the group has a strong preference for breast cancer, pick TNBC — but plan for the extra data wrangling up front.

At-a-Glance Comparison

	Melanoma	NSCLC	TNBC	CRC (MSI-H)	ccRCC
TCGA N	~473	~1,089	~170	~90–125*	~537
DepMap lines	~60	~150+	~55	~70+	~25–30
External (non-TCGA) cohorts	✅ Rich	✅ Rich	✅ Rich	✅ Good	✅ Good
Q1 response predictor	✅	✅	✅	✅	✅
Q2 cell viability	✅	✅	✅	✅	⚠️
Q3 ODE models	✅	⚠️	⚠️	✅	✅
Q4 drug targets	✅	✅	✅	✅	⚠️
Q5 chemo vs. immuno	✅	✅	✅	✅	✅
Overall	🏆 Best	👤 Size wins	👤 Breast pick	Solid 4th	Weakest Q2/Q4

*MSI-H subset only

Final Project: Cancer Selection Pitch

Goal: Pick one cancer (not a dataset) that gives us rich, multi-source data, a compelling treatment comparison story, and traction across at least 3 of the 5 project questions.

Project questions recap:

- **Q1** — Cross-dataset patient response predictor
- **Q2** — Cell viability predictor for standard-of-care drugs
- **Q3** — ODE models
- **Q4** — New drug targets via LINCS / DepMap
- **Q5** — Treatment type selection (e.g. chemo vs. immunotherapy)

The 5 Candidates

1. Melanoma (Skin Cutaneous Melanoma — SKCM)

Recommended pick

Melanoma is the textbook precision oncology cancer. The BRAF V600E/K mutation cleanly splits patients into three treatment arms, creating a natural framework for every project question.

Question	Fit	Notes
Q1	 Strong	TCGA + ICGC + multiple published IO trial cohorts with annotated response; cross-cohort generalisation is tractable
Q2	 Strong	~60 lines in DepMap/CCLL; vemurafenib, dabrafenib, trametinib, dacarbazine all in GDSC/PRISM
Q3	 Strong	Mature published ODE models for tumor-immune dynamics, T-cell infiltration, and BRAF inhibitor resistance
Q4	 Strong	Deep DepMap CRISPR data in melanoma lines; LINCS L1000 well-represented; BRAF-WT vs BRAF-mutant is a clean axis for dependency analysis
Q5	 Best	Three distinct arms: dacarbazine (chemo) / BRAF+MEK inhibitors (targeted) / PD-1 + CTLA-4 checkpoint inhibitors — biomarker-driven decision logic is clean and clinically established

Key cohorts:

Dataset	N	What it provides
TCGA-SKCM	~473	Expression, mutation, CNA, survival
ICGC Melanoma	~80	Independent validation
KEYNOTE-006 / CheckMate 066/067	~300–600 each	IO response labels (PD-1 / CTLA-4 arms)
DepMap CCLL	~60 cell lines	Drug sensitivity, CRISPR, expression

Why it wins: largest TCGA cohort of the focused cancer types, cleanest treatment decision logic, most mature ODE literature, and the most celebrated clinical story in modern oncology.

2. Non-Small Cell Lung Cancer (NSCLC — LUAD + LUSC)

Best for maximum sample size

NSCLC gives you the largest raw sample count of any option, with a strong immunotherapy story anchored to PD-L1 expression.

Question	Fit	Notes
Q1	✔ Strong	TCGA LUAD + LUSC combined; plus POPLAR, OAK, KEYNOTE-024 for IO response; TMB and PD-L1 as established predictive features
Q2	✔ Strong	~150+ cell lines in DepMap — largest of all candidates; cisplatin/carboplatin, paclitaxel, erlotinib, osimertinib all in GDSC/PRISM
Q3	⚠ Moderate	EGFR/ALK signalling ODEs exist; less unified than melanoma tumor-immune models
Q4	✔ Strong	Massive CRISPR dataset in DepMap; EGFR-mutant vs. KRAS-mutant vs. ALK-rearranged subgroups give multiple target discovery angles
Q5	✔ Strong	Platinum chemo / EGFR-targeted (osimertinib) / PD-1 checkpoint (pembrolizumab) — PD-L1 TPS score and EGFR mutation status drive the decision

Key cohorts:

Dataset	N	What it provides
TCGA-LUAD	~585	Adenocarcinoma; expression, mutation, survival
TCGA-LUSC	~504	Squamous cell; broadens treatment arms
POPLAR / OAK (atezolizumab)	~287 / ~850	IO response, PD-L1 status
KEYNOTE-024	~305	First-line pembrolizumab, PD-L1 ≥ 50%
DepMap CCLE	~150+ cell lines	Largest drug sensitivity resource of all 5 options

Caveats: LUAD and LUSC are biologically distinct subtypes being pooled — adds noise to response prediction unless you subtype. The "story" is slightly less clean than melanoma's.

3. Triple-Negative Breast Cancer (TNBC)

Best breast cancer subgroup; strong clinical urgency

TNBC is the highest-unmet-need breast cancer subtype: ER-/PR-/HER2-, no targeted therapy historically, and only recently gaining immunotherapy approvals. That makes the chemo-vs-immunotherapy question genuinely open and impactful.

Question	Fit	Notes
Q1	✔ Good	TCGA-BRCA TNBC + METABRIC + IMpassion130 + KEYNOTE-522; PD-L1 / TIL score / BRCA1/2 as candidate predictive features
Q2	✔ Good	~50–60 TNBC lines in DepMap (MDA-MB-231, HCC1937, MDA-MB-468); paclitaxel, carboplatin, olaparib, sacituzumab govitecan in GDSC/PRISM
Q3	⚠ Thinner	DNA damage / PARP repair ODEs exist; immune infiltration models less mature than melanoma
Q4	✔ Good	PARP, PI3K, EGFR, androgen receptor (LAR subtype) all viable discovery targets in DepMap
Q5	✔ Strong	Chemo (paclitaxel + carboplatin) vs. pembrolizumab + chemo (KEYNOTE-522 regimen) vs. olaparib (BRCA1/2-mutant) — three-arm comparison with a biomarker (PD-L1 / BRCA status) driving the split

Key cohorts:

Dataset	N	What it provides
TCGA-BRCA (TNBC)	~170	Expression, mutation, survival — small, needs supplementing
METABRIC (TNBC)	~300	Independent UK/Canada cohort
IMpassion130	~900	Atezolizumab + nab-paclitaxel; PD-L1 IC status
KEYNOTE-522	~1,174	Pembrolizumab + chemo neoadjuvant; pCR as endpoint
DepMap CCLE	~55 cell lines	Drug sensitivity, CRISPR

Caveats: TCGA alone is underpowered — data harmonisation across multiple cohorts is essential and adds 1–2 days of pipeline work. ODE question (Q3) is thinner.

4. Colorectal Cancer — MSI-H Focus (CRC)

Cleanest biomarker story after melanoma

CRC's MSI-high (microsatellite instable) vs. MSS (stable) split is one of oncology's clearest predictive biomarkers for immunotherapy response, mirroring melanoma's BRAF story.

Question	Fit	Notes
Q1	✓ Good	TCGA-COAD/READ total ~630; MSI-H subset ~90–125; KEYNOTE-177 as external IO validation cohort
Q2	✓ Good	~70+ CRC lines in DepMap; FOLFOX (5-FU + oxaliplatin), irinotecan, cetuximab (KRAS-WT), bevacizumab in GDSC/PRISM
Q3	✓ Good	Wnt/ β -catenin signalling ODEs well-published; EGFR-RAS-MAPK pathway models also available
Q4	✓ Good	RAS/RAF/PI3K dependencies in DepMap; KRAS G12C is an emerging target with LINCS perturbation data
Q5	✓ Strong	FOLFOX chemo / cetuximab targeted (KRAS-WT only) / pembrolizumab checkpoint (MSI-H only) — biomarker-stratified three-way split

Key cohorts:

Dataset	N	What it provides
TCGA-COAD	~461	Expression, mutation, MSI status
TCGA-READ	~166	Rectal cancer; can be merged or analysed separately
KEYNOTE-177	~307 MSI-H	Pembrolizumab vs. chemo; PFS endpoint
TCGA MSI-H subset	~90–125	Focused immunotherapy-relevant cohort
DepMap CCLE	~70+ cell lines	Decent drug sensitivity coverage

Caveats: MSI-H is only ~15–20% of CRC — your core immunotherapy-relevant cohort is small (~90–125 in TCGA). MSS CRC has very poor immunotherapy response, so you are effectively working with a cancer-within-a-cancer. Broadening to all CRC dilutes the story.

5. Clear Cell Renal Cell Carcinoma (ccRCC / KIRC)

Strong immunotherapy story; distinct mechanistic biology

ccRCC has one of the richest targeted-vs-immunotherapy comparison stories in oncology, built around VHL mutation and the switch from VEGF inhibitors to checkpoint immunotherapy.

Question	Fit	Notes
Q1	✓ Good	TCGA-KIRC ~537; CheckMate 025 / CheckMate 214 / JAVELIN Renal 101 provide IO response cohorts; IMDC risk score is a known predictor to benchmark against
Q2	⚠ Moderate	Fewer DepMap lines (~25–30); sunitinib, everolimus, pazopanib in GDSC — reasonable but not as dense as melanoma/NSCLC
Q3	✓ Good	HIF-1 α / VHL / VEGF pathway ODEs well-developed; angiogenesis models published
Q4	⚠ Moderate	VHL/mTOR/VEGFR dependency well-known (limits novelty); smaller CRISPR dataset in DepMap for RCC lines
Q5	✓ Strong	Sunitinib/pazopanib (VEGF-targeted) vs. nivolumab + ipilimumab (dual checkpoint) vs. everolimus (mTOR) — IMDC risk score biomarker drives the arm selection

Key cohorts:

Dataset	N	What it provides
TCGA-KIRC	~537	Expression, mutation, VHL status, survival
CheckMate 025	~821	Nivolumab vs. everolimus; OS endpoint
CheckMate 214	~1,096	Nivo + ipi vs. sunitinib; IMDC stratification
JAVELIN Renal 101	~886	Avelumab + axitinib; PD-L1 stratification
DepMap CCLE	~25–30 cell lines	Smallest cell line resource of the 5 candidates

Caveats: DepMap cell line coverage is the thinnest of all candidates, which weakens Q2 and Q4. VHL/mTOR biology is heavily studied, so drug target discovery (Q4) may yield limited novelty.

Rejected Cancers

These were considered and ruled out for the reasons below.

Cancer	Main reason for rejection
GBM (Glioblastoma)	Very poor immunotherapy response — Q5 collapses to essentially one arm (temozolomide chemo). Small TCGA-GBM cohort (~170). High-grade biology means survival endpoints are compressed.
Pancreatic Cancer (PAAD)	Limited immunotherapy response, no meaningful chemo-vs-immuno comparison for Q5. Small TCGA-PAAD cohort (~185). Modest DepMap representation. Stroma-heavy biology confounds cell line models.
AML / Leukemia	Rich drug sensitivity data in DepMap, but diverges from solid tumour ODE models (Q3). Different tissue microenvironment biology makes cross-question coherence harder. Treatment landscape (AML induction → transplant) doesn't map cleanly to Q5 framework.
Breast Cancer (all subtypes, undifferentiated)	Pooling all BRCA subtypes (Luminal A/B, HER2+, TNBC) obscures biology and weakens every question. Subgrouping is necessary, reducing effective N.
HER2+ Breast Cancer	Weaker than TNBC for this project specifically: limited chemo-vs-immunotherapy story (Q5), and "obvious" HER2 target reduces Q4 novelty. ODE models of HER2/EGFR signalling are strong, so if Q3 is the group's priority, HER2+ could be revisited.
Ovarian Cancer (HGSOC)	Good BRCA1/2 / PARP inhibitor story, but immunotherapy has had limited success, thinning Q5. Smaller TCGA cohort (~300 HGSOC). DepMap coverage modest.

Prepared for group discussion — data sources: TCGA, ICGC, DepMap/CCLE, GDSC, PRISM, published trial datasets.